Track 2 — Technical Documentation

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Track 2 — Technical Documentation

MVA Hackathon 2026 · Rare Disease, Real Kid · BUB1B / Mosaic Variegated Aneuploidy

This document describes how the Track-2 deliverables in this repository are produced and verified: the evidence workflow, the data sources and APIs it draws on, the native-image verification harness, the figure set, the check harness, and the candidate-table contract. It is the engineering companion to `TRACK2_REPORT.md`; it asserts no new science.

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1. Purpose and scope

Track 2 of the challenge asks for a mechanism-grounded drug-reprioritization dossier for the Track-1 diagnosis (biallelic BUB1B / Mosaic Variegated Aneuploidy). The deliverables in this repository are:

Artifact	Path	Role
Professional report	`TRACK2_REPORT.md`, `TRACK2_REPORT.pdf`	The submission document: mechanism characterization, therapeutic landscape, exclusion jurisprudence, six-assay validation plan
Lay report	`TRACK2_LAY_REPORT.md`, `.pdf`	Plain-language companion for the family/foundations audience
Candidate table	`candidates.tsv`	20-row machine-readable candidate register
Report figures	`figures/`	F1–F5 committed renders
Evidence pack	`evidence/`	19 analytical memos (manifest: `evidence/MEMO_MANIFEST.md`), four adversarial review passes, adjudication memo, structural validation
Native images	`evidence/native_images/`	Third-party structural imagery with a full verification chain (§4)
Citation check	`checks/check_citations.py`	Reference-consistency gate (§6)

Non-goals. Track-1 (variant calling/ranking) lives in its own repository. This document changes no scientific claim; where it quotes numbers, each is traceable to a repository artifact named next to it.

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2. Workflow architecture

The dossier was produced by a generation → adversarial-review → integrate loop, documented in `evidence/WORKFLOW_CONSENSUS.md` (the

session log) and closed per `evidence/SCOPE_DECISION.md`:

1. **Wave-2 axis generation** — parallel research memos per mechanism axis (e.g. `wave2_StabilizationAxis.md`, `wave2_SenolyticAxis.md`, `wave2_CINSyndromeAxis.md`, `wave2_NonsenseAxis.md`), each an evidence memo.
2. **Adversarial passes** — an out-of-family reviewer attacked each integrated pack (`wave2_ADVERSARIAL_REVIEW.md`, `wave3_ADVERSARIAL_PASS2.md`, `wave4_ADVERSARIAL_PASS3.md`, `wave5_ADVERSARIAL_PASS4.md`); findings were fixed at the controller layer. Major-count decay per `SCOPE_DECISION.md`: **8 → 2 → 2 → 0** across the four passes.
3. **Closure** — pass 4 returned zero majors; the stop rule (two consecutive zero-major passes, second over fresh axes) was declared met: *"CONVERGENCE OUTCOME (2026-09-05): decay 8 → 2 → 2 → 0; pass 4 returned ZERO majors; workflow CLOSED-WITH-RESIDUAL-RISK."* Residuals are declared in `SCOPE_DECISION.md` (R1: no pass 5 exists).
4. **Contested anchors** — claims a wave could not settle received dedicated adjudication memos (e.g. `adjudication_rapalog_contested_anchor.md`).

The consensus matrix and candidate register are the *output* of this loop: `candidates.tsv` carries a `consensus_status` column (INCLUDE / INCLUDE-AS-METABOLIC-CONTEXT / EXCLUDE …) that the report's §4 (therapeutic landscape) mirrors; the indicators documented there are what the six-assay validation plan (report §5) operationalizes.

Attack map. `evidence/TRACK2_ATTACK_MAP.md` lists, per report section, which claims were targeted and how each attack resolved — the audit surface for a judge.

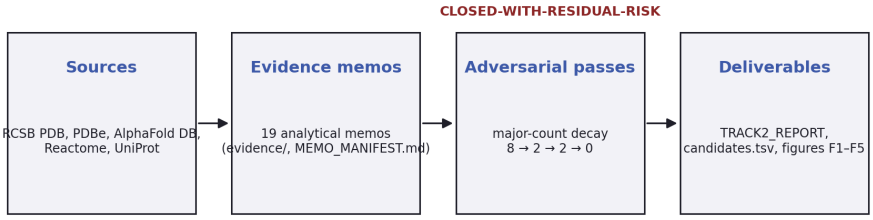


Figure 1: Workflow overview

3. Data sources and APIs

All external data enters through named, citable endpoints. Exact URL patterns live in `evidence/native_images/fetch_native.sh`.

Source	What we use	Endpoint pattern	Docs
Reactome	Native pathway diagrams (R-HSA-68877, R-HSA-69620, R-HSA-5663220); flagged export with `flg=O60566`; diagram layout JSON; participants membership	`ContentService/exporter/diagram/{stId}.png?download/current/diagram/{stId}.json, ContentService/data/participants/{stId}`	reactome.org/dev/content-service
RCSB PDB	Assembly images for 6JKK, 6JKM	`https://cdn.rcsb.org/images/structures/pdb_`	rcsb.org/docs/programmatic-access
PDBe	Deposited-chain statics for 6JKK, 6JKM	`https://www.ebi.ac.uk/pdbe/static/entry/{pdb}_deposited_ch`	ebi.ac.uk/pdbe
AlphaFold DB	AF-O60566-F1 model + per-residue confidence JSON	`https://alphafold.ebi.ac.uk/files/AF-O60566-F1-confidence_`	alphafold.ebi.ac.uk/api-docs
UniProt	O60566 (BUB1B) feature table cited in report captions	REST API	uniprot.org/help/api

Key facts established through these sources (all recorded in `evidence/native_images/PROVENANCE.md`):

- The Reactome exporter is **deterministic**: two unflagged fetches of the same diagram are byte-identical (negative control, 2026-09-24).

- The participants endpoint lists uniprot:**O60566** (BUB1B) as a component of **7 / 4 / 7 complexes** across R-HSA-68877 / R-HSA-5663220 / R-HSA-69620 respectively — because BUB1B is a kinetochore component in those diagrams, never a standalone node (except on R-HSA-69620, where the `BUB1B` particle, `hBUBR1:hBUB3:MAD2*:CDC20` MCC and `MCC:APC/C` are drawn explicitly).
- The core SAC sub-pathways (R-HSA-141430/141444/141405, R-HSA-69618) carry `hasDiagram=false`; R-HSA-69620 is the diagrammed parent covering the arm.

Analysis libraries: matplotlib, numpy Pillow, scipy.ndimage — no hand-rolled statistics (binomial test via `scipy.stats.binomtest`).

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4. Native-image verification chain

The structural imagery is not decorative: its claims are verified in three steps, each independently checkable from committed artifacts. The full text — including the negative control, the two *unexplained* R-HSA-68877 clusters, and the withdrawn earlier wording — is `evidence/native_images/PROVENANCE.md`.

1. **Fetch** (`fetch_native.sh`) — non-destructive by default (`OUT` override for refresh), records sources.
2. **Flag-vs-plain diff maps** — exported with `flg=O60566`; determinism control proves diffs are real render changes, not timestamps.
3. **Grounding** — (a) participants membership (7/4/7 complexes, peDbIds listed); (b) coincidence of diff clusters with the verified complex nodes, priced against a null baseline: 1000 random 150-unit boxes, seed 42 (`null_baseline.py`) — R-HSA-69620: **12/12** vs 9.9 % baseline, p ≈ 9e-13; R-HSA-5663220: **4/4** vs 38.8 %, p = 0.023; R-HSA-68877: **8/10** vs 58.6 %, p = 0.15 — underpowered there, so membership (3a) is the anchor on that diagram.
4. **Readable zoom set** — `make_readable.py` builds overview + zoom sheets (`readable_*_overview/zoom.png`); crops from diff clusters, node names from diagram-JSON intersection. All six committed PNGs are **byte-identical** to regeneration from the archived inputs (`python3 make_readable.py <stId> <tmpdir>`, compared with `cmp`).

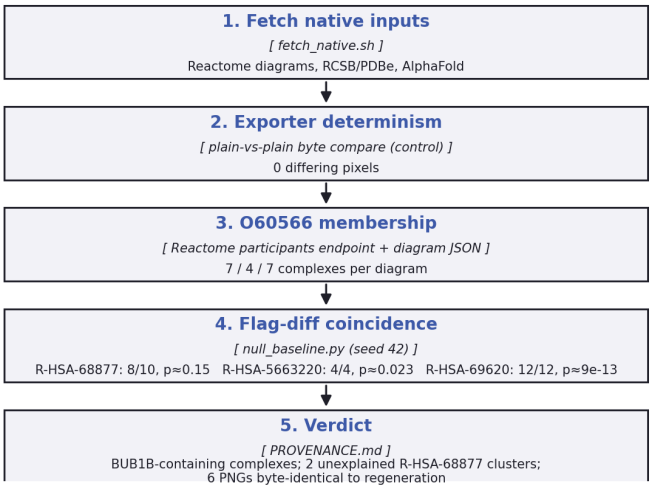


Figure 2: Verification chain

5. Report figure set F1–F5

Committed renders in `figures/`; the report's captions (§2–§5) name each figure's primary sources (UniProt O60566 feature table, Ensembl MANE transcript, AlphaFold model, 6JKK crystal structure).

Figure	Subject
F1_N1002K_structure.png	p.Asn1002Lys structural context (6JKK-based)
F2_BubR1_protein_map.png	BUBR1 functional-domain map with the two patient alleles
F3_BUB1B_gene_exon17.png	Exon-17 architecture for the ASO-skipping program
F4_decision_instrument.png	Gate instrument for therapeutic decisions
F5_consensus_process.png	The wave/consensus process summarized for the report

Related structural artifacts: `evidence/6JKK.cif`, `evidence/6JKK.pdb`, `evidence/AF_BUB1B.pdb`, `evidence/plip_6jkk/` (PLiP interaction profile outputs; PLiP = Protein–Ligand Interaction Profiler).

6. Citation check harness

`checks/check_citations.py` is the reference-consistency gate for the professional report:

```
python3 checks/check_citations.py $PWD    # run from anywhere; arg = repo root
```

It censuses the report's numbered reference block against in-text citations and prints `CENSUS OK` on success (current state: **61 base entries + 2 sub-rows, all cited**). A failing census blocks ship — run it after any edit to `TRACK2_REPORT.md`.

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7. Candidate-table contract

`candidates.tsv`: TSV, header + **20 data rows**.

Column	Content
`candidate`	Therapy candidate (generic name or class)
`axis`	Mechanistic axis letter (proteostasis, senolysis, immune surveillance, …)
`MoA`	Mechanism of action, one sentence
`approval_status`	Regulatory positioning (e.g. "investigational; no approved indication")
`evidence_grade`	Letter grade (A/B/C × human/preclinical)
`supporting_n`, `contrasting_n`	Study counts pro/contra
`key_PMIDs_DOIs`	Pinned primary sources, `PMID|DOI; …`
`consensus_status`	Wave-loop verdict (INCLUDE / INCLUDE-AS-… / EXCLUDE + reason)
`0..8`	Per-reviewer score rows from the consensus matrix

The TSV and report §4 vote from the same consensus matrix; divergence between them is a defect — the attack map tracks any found instance.

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8. Reproduction

```
# native-image inputs (non-destructive; use OUT=/tmp/x to refresh)
evidence/native_images/fetch_native.sh

# verification & derived figures
python3 evidence/native_images/null_baseline.py            # prints baselines + p-values
python3 evidence/native_images/make_readable.py R-HSA-68877 /tmp/out

# doc figures (this file's two diagrams)
python3 docs/figures/build_figs.py

# report consistency gate
python3 checks/check_citations.py $PWD

# PDF of this document (pandoc, xelatex)
docs/build_pdf.sh
```

pandoc manual governs PDF options; the two `docs/figures/*.png` are regenerated by `docs/figures/build_figs.py` (matplotlib, deterministic).

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9. References

- Reactome pathway database — Gillespie et al., 2022, PMID 34788843.
- AlphaFold Protein Structure Database / prediction — Jumper et al., *Nature* 2021, PMID 34265844.
- PLiP (Protein–Ligand Interaction Profiler) — Salentin et al., *Nucleic Acids Res.* 2015, PMID 26089450.
- RCSB Protein Data Bank — rcsb.org/docs/programmatic-access.
- PDBe — ebi.ac.uk/pdbe.
- UniProt — uniprot.org/help/api.
- matplotlib — Hunter, *Comput. Sci. Eng.* 2007, (matplotlib.org).
- Scientific-report citation block: see `TRACK2_REPORT.md` §8 (61+2 entries, registry-verified per §3 of that report).